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A model to evaluate the Human Error Probability in inspection tasks of a production system

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Abstract

Investigating the dynamics of human-machine interaction and its impacts on production performance is a key issue in the context of Industry 4.0 because the concept of the "operator 4.0", i.e., an operator integrated into a cyber-physical system, implies the need to manage complex human-machine systems. One of the most concerned fields on this topic is human reliability analysis, as the Human Error Probability (HEP) estimation by considering different work environment aspects. To this concern, the present work's purpose consists of assessing human error's impact on a manufacturing system by considering the different Performance Shaping Factors (PSFs) that affect the HEP. To this end, an analytical model has been developed to evaluate the human error in an inspection task to be accomplished in a full-real industrial case study. The HEP was estimated as a function of PSFs, including three different dimensions (i.e., task error proneness, operator's capabilities and characteristics of the work environment in the production system). It was found that the most impactful PSF affecting HEP depends on the working environment conditions in the production system. In this regard, the model shows that assuming an equal variation in the attributes related to all dimensions, the changes in working environment conditions from a physical and psychological point of view generate the most significant reduction in HEP. Consistently with these results, the average costs for reducing HEP by improving the working environment conditions are significantly higher than the average costs incurred to reduce the HEP considering the task error proneness and operator's capabilities.

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Keywords: human-machine systems, human error probability, human error cost, analytical model

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1. Introduction

Investigating the dynamics of human-machine interaction and its impacts on production performance is a key issue in the context of Industry 4.0 (I4.0). The adoption of the key enabling technologies of the fourth industrial revolution (e.g., Augmented Reality, Big Data and analytics, Additive Manufacturing, Industrial Internet & Cloud [1]) has become a flourishing reality thanks to the combination of production and network connectivity [2] that generate a symbiosis whereby the traditional role of the operator and of industrial processes changes. The transformation to smart processes in manufacturing leads to the higher deployment of data-driven approaches to support the worker [3]. To this concern, the view of the operator as cooperative working with automation systems [4] has been replaced by the concept of the "operator 4.0", i.e., an operator integrated into a cyber-physical system that enables him to exploit and enhance his individual skills in physical and cognitive terms [5]. If, on the one hand, employing technologies to enhance the inherent capabilities of the operator allows to improve the flexibility of the manufacturing systems [6]; on the other hand, it implies the need to manage complex human-machine systems (HMSs) [7]. The operator constantly interacts with advanced technological systems in an innovative production system. One of the most affected fields by this industrial transition is the Human Reliability Analysis (HRA), aiming to evaluate the Human Error Probability (HEP) in a task accomplishment. Consistent with this end, it is needed to identify the different factors affecting the operator's behaviour in completing a task, considering all the elements of complexity included in the cyber-physical system from a physical and cognitive point of view. Evidence in scientific literature indicates that errors committed by operators are responsible for 60–90% of accidents [8]. To this concern, several methods have been developed to predict the HEP, considering both the physical and cognitive aspects.

The Cognitive Reliability and Error Analysis Method (CREAM) can be used according to a predictive approach to forecast the potential human error, and to retrospectively approach, to analyse and evaluate the human error [9]. The Human Error Assessment and Reduction Technique (HEART) method identifies the different conditions leading to an increase in the HEP [10]. Similarly, the Technique for Human Error Rate Prediction (THERP) evaluates the HEP using a fault-tree analysis according to a top-down approach to identify the component level failures (basic event) that cause the system level failure (top event) to occur [11].

The Simulator for Human Error Probability Analysis (SHERPA) model combines the advantages of the simulation tools and the traditional HRA methods to identify human behaviour and predict the HEP for a given scenario in complex industrial systems [12]. Although all the mentioned models have their peculiarities in the estimation of HEP, they all have in common an aspect; in all cases, the HEP estimation is conducted starting from a set of factors affecting the likelihood of a human error occurring. These factors are generally referred Performance Shaping Factors (PSFs) [13]. It is noteworthy that the HEP estimation is a multi-dimensional problem, depending on internal and external systems factors. External factors refer to work environment conditions. The internal factors refer to human factors related to the psychological and physiological conditions of the operators [14]. Therefore, for the HEP estimation in a given situation, it is needed to investigate the influence that each PSF has on the likelihood of a human error occurring, identifying the main causes of error and the main aspects to be changed to improve the performance of the operator and, consequently, of the production system. In general cases, actions allowing to reduce the HEP represent an additional cost to be sustained. Therefore, the analysis of the factors affecting HEP by evaluating the associated costs leads to planning the resources to be invested in improving the system in a more efficient and effective way.

Nevertheless, most of the research works available in the scientific literature identify PSFs and estimate human reliability, neglecting the cost due to the strategies to be implemented to reduce the probability of human error. In this regard, therefore, a research gap arises, which this paper aims to fill. Therefore, the present work's purpose consists of assessing human error's impact on a manufacturing system by considering the different PSFs that affect the HEP, starting from an existing methodology. To this end, an analytical model has been developed to evaluate the human error in tasks to be accomplished in a full-real industrial case study. Aiming to fill the research gap that emerged, a sensitivity analysis has allowed identifying the most impactful aspects of HEP from an economic perspective.

The rest of the paper is organised as follows: section 2 illustrates the analytical model developed to evaluate the cost of reducing human error by performing a task, and section 3 describes the full-real industrial case study where to model was applied with respect to the sensitivity analysis. In section 4, the results achieved are shown and discussed and finally, in section 5, the conclusions of the present work and suggestions for future studies are provided.

2. Materials and methods

The objective of the present work consists of assessing human error's impact on a manufacturing system by considering the different PSFs that influence the HEP. To this end, an analytical model has been developed for evaluating the human error reduction cost ($C_{red_{HE}}$). It has been defined as the required cost to reduce the HEP from a current scenario (identified with 0), at which a HEP_0 value was evaluated, to an improved scenario (identified with 1), where the HEP was reduced, such that HEP_1<HEP_0. The outline of the methodology adopted is summarized in Figure 1.

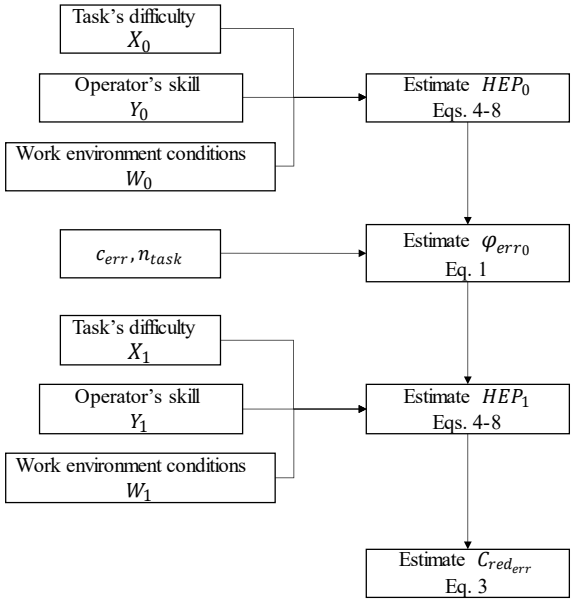


Fig. 1. Outline of the methodology employed

A total human error cost function has been identified for the current (φ_{err_0}) and improved scenario (φ_{err1}), according to equations 1 and 2, respectively.

$$\varphi_{err_0}\left[\frac{€}{y}\right] = HEP_0 \cdot c_{err} \cdot n_{task} \tag{1}$$

$$\varphi_{err_1}\left[\frac{€}{y}\right] = C_{red_{HE}} + HEP_1 \cdot c_{err} \cdot n_{task} \tag{2}$$

Where c_{err} $[€/task]$ is a penalty cost due to incorrect task execution, n_{task} $[task/y]$ identifies the number of repetitions of the task to be performed in a year by an operator, and $C_{red_{HE}}$ $[€/y]$ is the cost needed to reduce the HEP's value from current (HEP_0) to improved (HEP_1) scenario. The minimum human error reduction cost ($C_{red_{err}}$) was evaluated according to equation 3.

$$C_{red_{err}}(HEP) = (HEP_0 - HEP_1) \cdot c_{err} \cdot n_{task} \tag{3}$$

The evaluation of HEP has been carried out according to the model introduced in 2008 by Elmaraghy *et al*. [15]. In this case, the HEP in the execution of a task depends on the task error proneness, the capabilities of the operator and on the conditions of the working environment, according to equation 4:

$$HEP_i = \left(u_i(X_i, Y_i, Z_i)\right)^6, i = 0{,}1 \tag{4}$$

Where u_i represents the utility function referred to different PSFs depending on a set of attributes $X_i = \{x_{1i}, x_{2i}, x_{3i}, x_{4i}, x_{5i}\}$, $Y_i = \{y_{1i}, y_{2i}\}$, and $Z_i = \{z_{1i}, z_{2i}, z_{3i}, z_{4i}\}$, related to the task error proneness, operator's capabilities, and work environment conditions, respectively. The utility function is expressed in accordance with equations 5 and 6:

$$u_i(X_i, Y_i, Z_i) = w_{X_i} u_{X_i} + w_{Y_i} u_{Y_i} + w_{Z_i} u_{Z_i} \tag{5}$$

$$w_{X_i} + w_{Y_i} + w_{Z_i} = 1 \tag{6}$$

Where $u_{X_i}, u_{Y_i}, u_{Z_i}$ are the utility values related to the attribute sets X_i, Y_i and Z_i, respectively, and $w_{X_i}, w_{Y_i}, w_{Z_i}$ are the relative weights (Eqs. 7 and 8).

$$u_{X_i} = \sum_{j=1}^{5} w_{x_{j,i}} u_x(x_{j,i}) \tag{7}$$

$$\sum_{j=1}^{5} w_{x_{j,i}} = 1 \tag{8}$$

The set of attributes included in X_i, Y_i and Z_i were identified according to the task's features, the operator's skill, and the work environment. In particular, $x_{1,i}$ and $x_{2,i}$ are the attributes related to the average difficulty of the task under cognitive and physical perspectives, respectively. The values were subjectively assessed in both cases according to a scale from 0 to 10, where 0 identifies the easiest and 10 for the most difficult. $x_{3,i}$ identifies the percentage of diversity between the task elements, $x_{4,i}$ represents the coupling degree between the task elements, and $x_{5,i}$ is the level of implementation of tools and techniques to support the task and reduce error.

Similarly, the attribute $y_{1,i}$ identifies the operator's capabilities from a personal point of view (e.g., attitude, motivation, age, culture), $y_{2,i}$ provides an evaluation of the operator's capabilities from a professional point of view (e.g., experience. Both cases' values were subjectively assessed on a scale from 0 to 10.

The work environment's attributes $z_{1,i}$ and $z_{2,i}$ represents the adequacy of the working environment from an ergonomic and psychological point of view, respectively; the values of these variables were subjectively assigned with a score from 0 to 10. The attribute $z_{3,i}$ identifies the time pressure available to perform the task, given by the rate between the operator's task time and the planned cycle time; $z_{4,i}$ depends on the reconfiguration frequency of the production system. It is estimated by assigning a null value at no reconfiguration of the production system in 10 years and a unit value at a reconfiguration frequency of one per shift.

3. Full-real industrial case

The developed model has been applied to the real industrial case of a large manufacturing company producing electronic boards for industrial applications. The plant is located in the south part of Italy. The model was tested to evaluate an inspection task in an electronic board assembly process. It consists of the operator's visual check of the correct positioning of the components on the printed circuit board previously carried out by the pick and place machine. The operation is assisted by an automated optical inspection machine equipped with a camera system and a pre-loaded programme with the correct board model. The machine supports the operator in identifying defects such as missing components, positioning error, polarity inversion, low quantity of solder paste, short circuit between two adjacent pins, etc. The task is performed correctly if the operator can detect defects on the circuit board without providing false alarms. In other words, there is a failure on the inspection task if the operator identifies a defect on a good circuit board or, in the alternative, if he does not detect any defect on a damaged circuit board.

Table 1 shows the decomposition of the task into elementary operations according to a classification in physical (Mp_i) and cognitive (Mc_j) sub-tasks.

Table 1. Decomposition of the analysed task into elementary operations.

Task element	Operation description
Mp_1	Take the mounted boards from the pick and place station.
Mp_2	Look at the processing sheet.
Mp_3	Place the board on the automated optical inspection machine.
Mc_1	Program the machine with the correct board model.
Mp_4	Look at the board on the machine.
Mc_2	Correct errors marked by the machine in case they are false.
Mc_3	Decide whether the board is good or defective.
Mp_5	Unload the boards from the rack and place them downstream of the automated optical inspection process.

The subjective assessment of the inherent difficulty of physical (Dp_i) and cognitive (Dc_j) task elements and their average values are shown in Table 2.

Table 2. Assessment of inherent difficulty of physical and cognitive task elements.

	Mp_1	Mp_2	Mp_3	Mc_1	Mp_4	Mc_2	Mc_3	Mp_5	Average
Dp_j	2	3	4	-	4	-	-	2	3
Dc_j	-	-	-	4	-	4	4	-	4

A matrix has been constructed to assess the coupling between task elements (Table 3). Coupling between two task elements implies that if one of the task elements is not performed correctly, the other task element is expected to be affected. The following task elements are coupled Mp_2-Mc_1, Mp_3-Mc_1, Mp_4-Mc_1, Mp_4-Mc_2, and Mc_2-Mc_3.

Table 3. Task elements coupling matrix.

	Mp_1	Mp_2	Mp_3	Mp_4	Mp_5	Mc_1	Mc_2	Mc_3
Mp_1	1	-	-	-	-	-	-	-
Mp_2	-	1	-	-	-	1	-	-
Mp_3	-	-	1	-	-	1	-	-
Mp_4	-	-	-	1	-	1	1	-
Mp_5	-	-	-	-	1	-	-	-
Mc_1	-	1	1	1	-	1	-	-
Mc_2	-	-	-	1	-	-	1	1
Mc_3	-	-	-	-	-	-	1	1

According to section 2, therefore, Table 4 resumes the values adopted to evaluate the HEP_0 value in the current scenario and the corresponding total human error cost function (φ_{err_0} [€/y]). The relative weight of the attributes has been adopted according to [15].

Table 4. Values of the variables employed to evaluate HEP_0 and φ_{err_0} in the case study situation.

Weights	Value	Attributes	Value	Other variables	Value
w_{X_0}	0.428	$x_{1,0}$	4	HEP_0	0.001968585

w_{Y_0}	0.143	$x_{2,0}$	3	c_{err}	3000.102
w_{Z_0}	0.429	$x_{3,0}$	0.625	n_{task}	12750
$w_{x_{1,0}}$	0.221	$x_{4,0}$	0.179	φ_{err_0}	75300.95
$w_{x_{2,0}}$	0.046	$x_{5,0}$	0		
$w_{x_{3,0}}$	0.221	$y_{1,0}$	7		
$w_{x_{4,0}}$	0.122	$y_{2,0}$	8		
$w_{x_{5,0}}$	0.39	$z_{1,0}$	7		
$w_{y_{1,0}}$	0.333	$z_{2,0}$	7		
$w_{y_{2,0}}$	0.667	$z_{3,0}$	0.75		
$w_{z_{1,0}}$	0.109	$z_{4,0}$	0		
$w_{z_{2,0}}$	0.189				
$w_{z_{3,0}}$	0.351				
$w_{z_{4,0}}$	0.351				

A sensitivity analysis has been conducted to identify the influence of PSFs on the value of HEP. Consistent with this end, the values assigned to the set of attributes in X_i, Y_i and Z_i were changed, aiming to improve the current scenario. In particular, it was assumed a higher level of additional tools' implementation and the adoption of better techniques to support the tasks and reduce errors in cases #1, #4, and #7. The implementation level to support the task was increased by 10%, 20% and 30%, respectively. The other PSFs related to the operator's capabilities and work environment conditions were held constant.

Similarly, have been assumed more skilled operators in cases #2, #5, and #8, improving their capabilities by 10%, 20% and 30%, respectively. No improvement in terms of techniques to support the tasks and work environment conditions was considered in these cases.

More comfortable work environment conditions were assumed in cases #3, #6, and #9, improving the adequacy of the working environment from an ergonomic and psychological point of view, on the assumption that these changes lead to improve work environment conditions of 10%, 20% and 30%, respectively. No improvement in terms of techniques to support the task and operator's capabilities was considered in these cases. The attribute values identified for each case were summarized in table 5.

Table 5. Attributes' values are employed in the sensitivity analysis.

Improvement rate		10%			20%			30%	
Attribute	#1	#2	#3	#4	#5	#6	#7	#8	#9
$x_{1,1}$	3.6	4	4	3.2	4	4	2.8	4	4
$x_{2,1}$	2.7	3	3	2.4	3	3	2.1	3	3
$x_{3,1}$	0.5625	0.625	0.625	0.5	0.625	0.625	0.4375	0.625	0.625
$x_{4,1}$	0.1611	0.179	0.179	0.1432	0.179	0.179	0.1253	0.179	0.179
$x_{5,1}$	0	0	0	0	0	0	0	0	0
$y_{1,1}$	7	7.7	7	7	8.4	7	7	9.1	7
$y_{2,1}$	8	8.8	8	8	9.6	8	8	10.4	8
$z_{1,1}$	7	7	7.7	7	7	8.4	7	7	9.1
$z_{2,1}$	7	7	7.7	7	7	8.4	7	7	9.1
$z_{3,1}$	0.75	0.75	0.675	0.75	0.75	0.6	0.75	0.75	0.525
$z_{4,1}$	0	0	0	0	0	0	0	0	0

It should be noted that the attribute value has been increased or decreased depending on the trend of the utility function considered, in accordance with [15].

4. Results and discussions

The results achieved from the model implementation according to assumptions described in the previous section are shown in Figure 2.

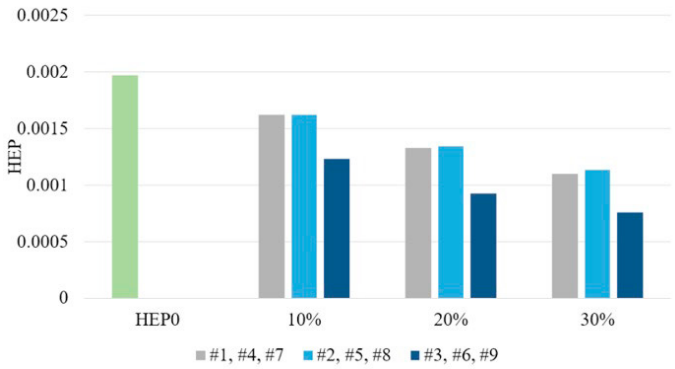


Fig. 2. HEP1 was evaluated by varying PSFs in the industrial case study.

As can be observed, there is a decrease in HEP in all scenarios with respect to the HEP_0 value, approximately equal to 0.2%. Regarding the extent of HEP reduction evaluated, it can be observed that the values identified in cases #1 and #2, #4 and #5, #7 and #8, respectively, are very similar. This implies that by varying the values of the attributes of the X_i and Y_i, a similar change in HEP is expected. In cases #1 and #2, there is a reduction of 17.86% and 17.77% with respect to the value of HEP_0. In cases #4 and #5 there is a reduction of 32.39% and 31.81%; similarly in cases #7 and #8 of 44% and 42.5%. Therefore, the PSFs related to the task error proneness and operator's capabilities have a comparable significance concerning the impact of HEP.

The HEP estimation is quite different for cases where an improvement in work environment conditions was assumed. Consistent with this consideration, in cases #3, #6 and #9, a reduction of HEP value by 37.65%, 53.16% and 61.63% has been identified compared to HEP_0. This implies that the attributes that most significantly influence the HEP are related to the work environment conditions of the production systems.

In all cases, the minimum human error reduction cost ($C_{red_{err}}$) has been identified (fig. 3). As it can be observed, the $C_{red_{err}}$ are consistent with the expected changes of HEP.

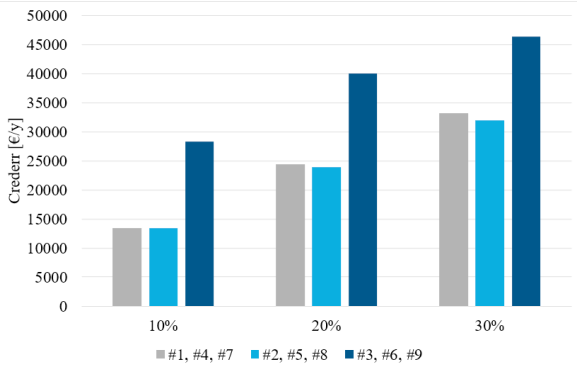


Fig. 3. $C_{red_{err}}$ was evaluated by varying PSFs in the industrial case study.

Therefore, the human error reduction costs are comparable when the improvements are focused on aspects related to task error proneness (X_i) and operator skills (Y_i). At the same time, they are significantly higher in case improvements in the production system's work environment conditions (Z_i) are needed. To this concern, the cost identified in cases #1 and #2, #4 and #5, #7 and #8, respectively, are very similar. Indeed, in cases, #1 and #2 a $C_{red_{err}}$ of 13449.32 [€/y] and 13380.07 [€/y] were identified in cases #4 and #5 a value of 24392.47 [€/y] and 23956.9 [€/y] is identified, and, finally, in cases #7 and #8, the cost of human error reduction is 33184.78 [€/y] and 32005.01 [€/y], respectively. Consistent with the changes of HEP according to aspects to be improved, the costs of reducing human error are significantly higher when acting on the set of attributes Z_i. In cases #3, #6 and #9, indeed $C_{red_{err}}$ is equal to 28347.44 [€/y], 40028.73 [€/y], and 46407.56 [€/y], respectively, values that are approximately 111%, 65% and 42% higher than the costs needed to improve aspects focused on task error proneness and operator skills.

5. Conclusions

The present work's purpose consisted of assessing human error's impact on a manufacturing system by considering the different PSFs affecting the HEP. To this end, an analytical model has been developed based on a HEP estimation methodology that considers the PSFs depending on three aspects, i.e., the task error proneness, the operator's capabilities and the working environment conditions in the production systems. The model has been applied to a real case study of a manufacturing industry producing electronic boards, in particular to a visual inspection task in the electronic board assembly process. Starting from the HEP value identified in the case study, a sensitivity analysis has been performed to evaluate the influence of the PSFs considered on the reduction of human error and, consequently, on the cost to be incurred. The results proved that the production system's working environment conditions are the most impactful in reducing the HEP. In this regard, it has been found that with the same changes in percentage terms of the attributes related to all PSFs, the change in working environment conditions from a physical and psychological point of view generates the most significant reduction in HEP. Consistent with this result, it was identified that the corresponding average costs to improve the aspects related to work environments are 111%, 65% and 42% higher than the average costs incurred by varying the attributes of the other two categories by 10%, 20% and 30%, respectively.

The developed model may be helpful for several purposes in industrial contexts. First, it provides a preliminary economic evaluation of human error's effects on production system performance, correlating the HEP with the cost incurred for production losses. Secondly, it allows highlighting the analysed tasks' main criticalities, leading to potential errors. Finally, in a business context, the model supports the decision maker in choosing the best strategy between possible alternatives to reduce the HEP, maximizing the performance from an economic perspective.

The model can be adopted in any work environment. It is possible to scale the model by correlating to each PSF considering the cost of error reduction so that a cost-effective combination of improvement actions can also be evaluated. Although the adoption of the developed model helps identify the best strategy to reduce the HEP considering the performance and economic issues, the present work shows limitations that can be overcome in future studies. First, the assessment of the cost of reducing human error refers to the category of attributes and not to the

individual attribute. In this regard, the present model can be deepened by assessing the costs associated with each attribute considered in the model.

Moreover, in the present work, there is no optimisation of the error reduction cost by considering a mix of attributes belonging to different categories. In this regard, future studies may exploit the developed total error cost function to minimize the HEP considering constraints related to investment availability and production environment conditions. A further limitation of the proposed methodology is the operator's subjective task difficulty assessment. This issue could be improved by replacing the subjective assessment by the operator with the assessment by a panel of experts or by adopting statistically relevant data identified for each business sector, available in the scientific literature (e.g., Core-Data [16], offshore HEP data [17], HEP for railway operations [18], etc.).

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